AS1109 - Extended Solved Practice Set II

1. Squeeze theorem with an oscillating factor

Evaluate

$$\lim_{(x,y)\to(0,0)} (x^2+y^2)\cos\left(\frac{1}{\sqrt{x^2+y^2}}\right).$$

Solution

Since $|\cos(\cdot)| \le 1$,

$$\left|(x^2+y^2)\cos\left(\frac{1}{\sqrt{x^2+y^2}}\right)\right| \le x^2+y^2 \to 0.$$

Hence

$$\boxed{0}.$$

2. A limit that depends on direction

Determine whether

$$\lim_{(x,y)\to(0,0)} \frac{(x+y)^2}{x^2+y^2}$$

exists.

Solution

Along $y=0$ the expression tends to 1. Along $y=-x$ it equals 0. Therefore

$$\boxed{\text{the limit does not exist}}.$$

3. Continuity extension with a logarithm

For $(x,y) \neq (0,0)$ let

$$f(x,y) = \frac{\ln(1+x^2+y^2)}{x^2+y^2}.$$

Choose $f(0,0)$ so that f is continuous at the origin.

Solution

Put $u = x^2+y^2$. Then

$$\lim_{u\to 0} \frac{\ln(1+u)}{u} = 1.$$

Therefore

$$\boxed{f(0,0)=1}.$$

4. Continuity proved by an estimate

Show that

$$h(x,y) = \begin{cases} \dfrac{xy^2}{x^2+y^2}, & (x,y)\neq(0,0), \\ 0, & (x,y)=(0,0) \end{cases}$$

is continuous at $(0,0)$.

Solution

Since $y^2 \le x^2+y^2$,

$$|h(x,y)| = |x|\frac{y^2}{x^2+y^2} \le |x| \to 0.$$

Thus

$$\boxed{h \text{ is continuous at } (0,0)}.$$

5. Domain and range of a logarithmic function

Find the domain and range of

$$f(x, y) = \ln(y - x^2),$$

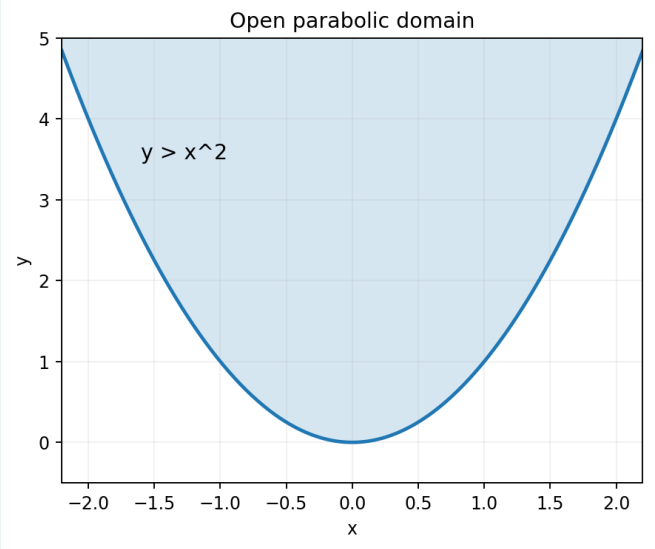
and classify the domain.

Solution

$$y - x^2 > 0 \quad \Longrightarrow \quad \boxed{y > x^2}.$$

The domain is **open and unbounded**. Since $y - x^2$ can take every positive value,

$$\boxed{\text{Range}(f) = \mathbb{R}}.$$



6. A bounded region that is neither open nor closed

Classify

$$S = \{(x, y) : x^2 + y^2 < 9,\ y \geq 0\}.$$

Solution

The circular arc is excluded, while the diameter $y = 0$ is included. Hence the set is neither open nor closed. It lies inside a radius-3 disk, so it is bounded:

$$\boxed{S \text{ is neither open nor closed, and it is bounded}}.$$

7. Level curves that are parabolas

For

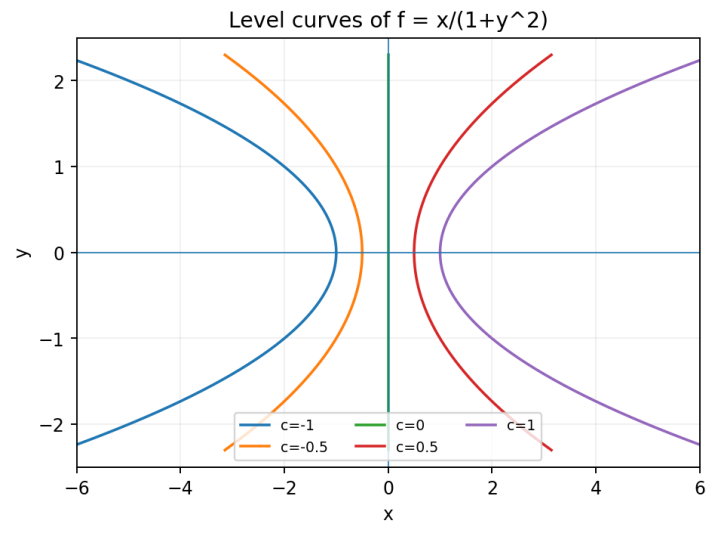
$$f(x, y) = \frac{x}{1 + y^2},$$

find the level curves $f(x, y) = c$.

Solution

$$\frac{x}{1 + y^2} = c \quad \Longrightarrow \quad \boxed{x = c(1 + y^2)}.$$

For $c > 0$ they open right, for $c < 0$ they open left, and for $c = 0$ the level curve is $x = 0$.



8. Inverse and composition

Let

$$p(t) = t^3 + 1, \qquad q(t) = \frac{1}{t+2}.$$

Find p^{-1}, $(p \circ q)(t)$ and $(q \circ p)(t)$, including restrictions.

Solution

$$\boxed{p^{-1}(s) = \sqrt[3]{s-1}},$$

$$\boxed{(p \circ q)(t) = \frac{1}{(t+2)^3} + 1, \quad t \neq -2},$$

$$\boxed{(q \circ p)(t) = \frac{1}{t^3+3}, \quad t \neq -\sqrt[3]{3}}.$$

9. Traces and contours of a non-polynomial surface

For

$$z = e^x + y^2,$$

find the traces in $y = 0$ and $x = 0$, and describe the level curves $z = c$.

Solution

At $y = 0$,

$$\boxed{z = e^x}.$$

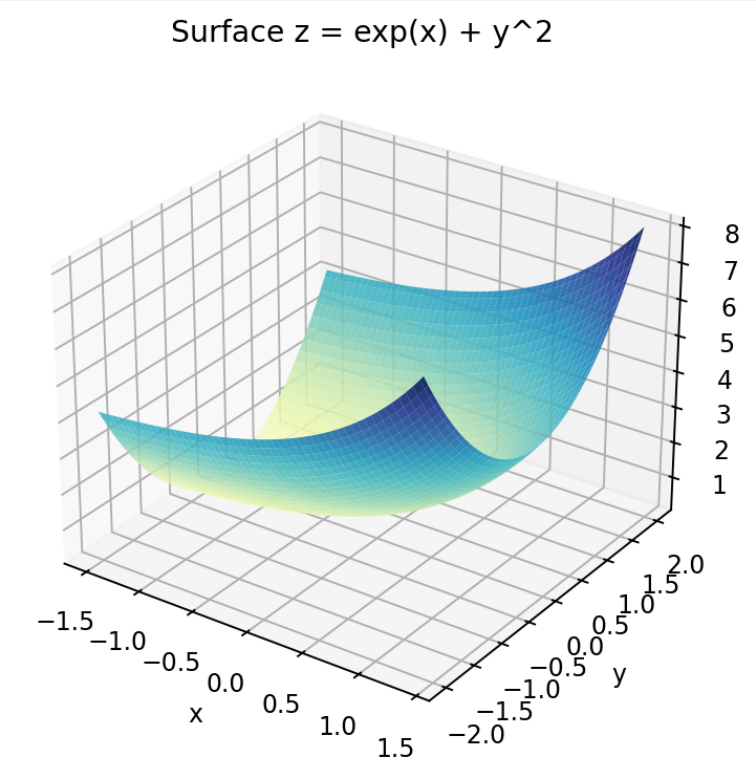
At $x = 0$,

$$\boxed{z = 1 + y^2}.$$

For $z = c$,

$$e^x + y^2 = c \quad \Longrightarrow \quad \boxed{y^2 = c - e^x},$$

with real points only where $e^x \leq c$.



10. Partial derivatives with several rules

For

$$f(x,y) = x^2 \cos y + ye^x,$$

find f_x, f_y, f_{xx}, f_{yy} and f_{xy}.

Solution

$$f_x = 2x\cos y + ye^x, \qquad f_y = -x^2 \sin y + e^x,$$
$$f_{xx} = 2\cos y + ye^x, \qquad f_{yy} = -x^2 \cos y,$$
$$\boxed{f_{xy} = f_{yx} = -2x\sin y + e^x}.$$

11. Implicit partial differentiation

The equation

$$x^2 z + ye^z = 3$$

defines $z = z(x,y)$ near $(1,0,3)$. Find z_x and z_y there.

Solution

Let $F = x^2 z + ye^z - 3$. Then

$$F_x = 2xz, \qquad F_y = e^z, \qquad F_z = x^2 + ye^z.$$

Hence

$$z_x = -\frac{F_x}{F_z}, \qquad z_y = -\frac{F_y}{F_z}.$$

At $(1,0,3)$,

$$\boxed{z_x = -6, \qquad z_y = -e^3}.$$

12. Hessian matrix

Find the Hessian of

$$f(x,y) = \sin(x+y) + x^2 y^2.$$

Solution

$$f_{xx} = -\sin(x+y) + 2y^2,$$
$$f_{yy} = -\sin(x+y) + 2x^2,$$
$$f_{xy} = f_{yx} = -\sin(x+y) + 4xy.$$

Thus

$$\boxed{H_f = \begin{pmatrix} -\sin(x+y) + 2y^2 & -\sin(x+y) + 4xy \\ -\sin(x+y) + 4xy & -\sin(x+y) + 2x^2 \end{pmatrix}}.$$

13. Partial-differential-equation check

Verify that

$$u(x,t) = e^{x+2t}$$

satisfies

$$u_{tt} = 4u_{xx}.$$

Solution

$$u_{xx} = e^{x+2t}, \qquad u_{tt} = 4e^{x+2t}.$$

Therefore

$$\boxed{u_{tt} = 4u_{xx}}.$$

14. Total differential

For

$$z = \ln(x^2 + y^2),$$

estimate the change in z from $(3,4)$ when

$$dx = 0.02, \qquad dy = -0.01.$$

Solution

At $(3,4)$,

$$z_x = \frac{6}{25}, \qquad z_y = \frac{8}{25}.$$

Hence

$$dz = z_x dx + z_y dy = \frac{6}{25}(0.02) + \frac{8}{25}(-0.01) = \boxed{0.0016}.$$

15. Linearization

Find the linearization of

$$f(x,y) = e^x \sqrt{y}$$

at $(0,4)$ and use it to approximate $f(0.01, 4.04)$.

Solution

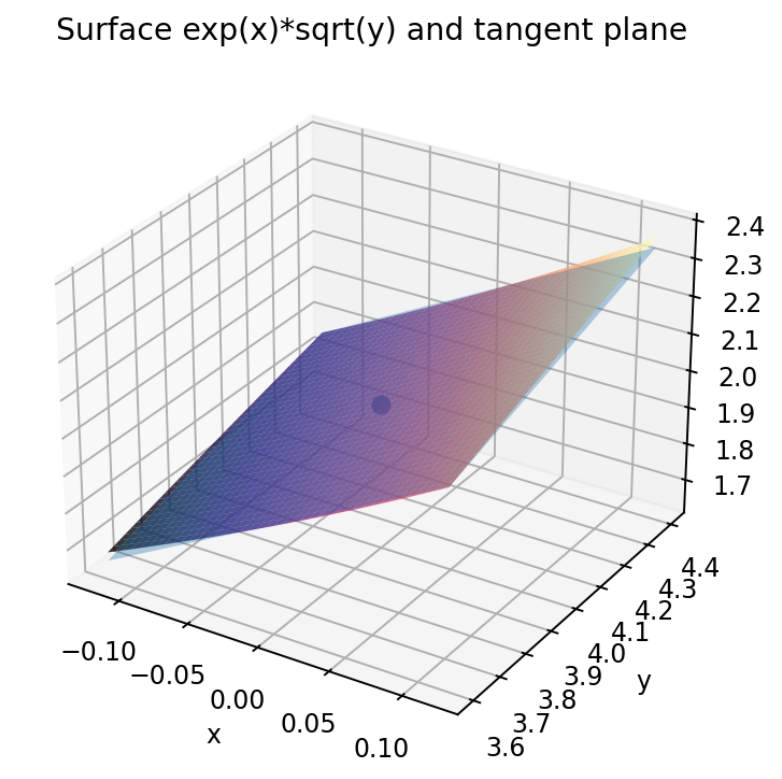
$$f(0,4) = 2, \qquad f_x(0,4) = 2, \qquad f_y(0,4) = \frac{1}{4}.$$

Thus

$$\boxed{L(x,y) = 2 + 2x + \frac{1}{4}(y-4)}.$$

Therefore

$$f(0.01, 4.04) \approx 2 + 0.02 + 0.01 = \boxed{2.03}.$$



16. Ideal-gas sensitivity

For fixed n,

$$V = \frac{nRT}{P}.$$

Estimate the percentage change in V if temperature rises by 1.2% while pressure falls by 0.8%.

Solution

$$\frac{dV}{V} = \frac{dT}{T} - \frac{dP}{P} = 1.2\% - (-0.8\%) = 2.0\%.$$

Hence

$$\boxed{V \text{ increases by approximately } 2.0\%.}$$

17. Natural-frequency sensitivity

A mass-spring model has

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}.$$

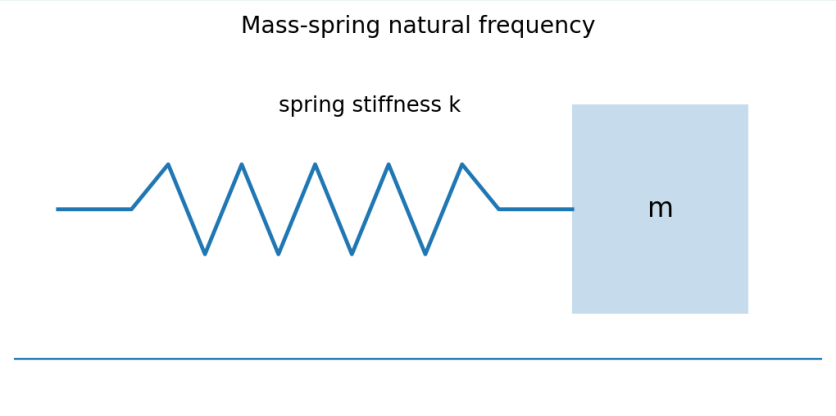
Estimate the percentage change in f if k increases by 2% and m increases by 1%.

Solution

$$\frac{df}{f} = \frac{1}{2}\frac{dk}{k} - \frac{1}{2}\frac{dm}{m} = 1\% - 0.5\% = 0.5\%.$$

Thus

$$\boxed{f \text{ increases by approximately } 0.5\%.}$$



18. Two-variable chain rule

Let

$$z = xy + y^2, \qquad x = u^2 - v, \qquad y = u + v^2.$$

Find z_u and z_v, and evaluate at $(u, v) = (1, 0)$.

Solution

$$z_x = y, \qquad z_y = x + 2y.$$

Also

$$x_u = 2u,\ y_u = 1, \qquad x_v = -1,\ y_v = 2v.$$

Therefore

$$z_u = 2uy + x + 2y, \qquad z_v = -y + 2v(x + 2y).$$

At $(u, v) = (1, 0)$, $x = y = 1$, so

$$\boxed{z_u = 5, \qquad z_v = -1}.$$

19. One-parameter chain rule

Let

$$W = x^2 y, \qquad x = e^t, \qquad y = 1 + t^2.$$

Find dW/dt at $t = 0$.

Solution

$$W_x = 2xy, \qquad W_y = x^2,$$
$$x' = e^t, \qquad y' = 2t.$$

Thus

$$\frac{dW}{dt} = W_x x' + W_y y'.$$

At $t = 0$, $x = y = 1$, $x' = 1$, $y' = 0$, hence

$$\boxed{\left.\frac{dW}{dt}\right|_{t=0} = 2}.$$

20. Three critical points

Find and classify the critical points of

$$f(x, y) = x^4 + y^4 - 4xy.$$

Solution

$$f_x = 4x^3 - 4y, \qquad f_y = 4y^3 - 4x.$$

Thus $y = x^3$ and $x = y^3$, giving

$$(0,0), \qquad (1,1), \qquad (-1,-1).$$

Also

$$f_{xx} = 12x^2, \qquad f_{yy} = 12y^2, \qquad f_{xy} = -4.$$

At $(0,0)$,

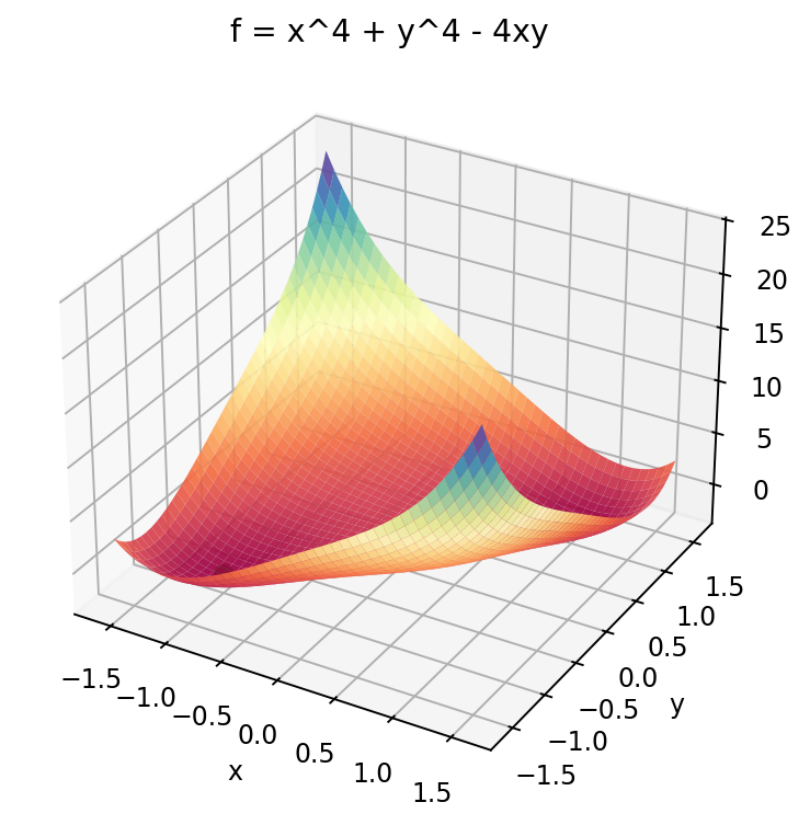
$$D = -16 < 0,$$

so it is a saddle. At $(1,1)$ and $(-1,-1)$,

$$D = 128 > 0, \qquad f_{xx} = 12 > 0,$$

so both are local minima.

$$\boxed{(0,0) \text{ saddle}; \quad (1,1), (-1,-1) \text{ local minima.}}$$



21. Absolute extrema on a closed disk

Find the absolute extrema of

$$f(x,y) = x^2 + y^2 + x$$

on

$$x^2 + y^2 \le 1.$$

Solution

Interior critical point:

$$2x + 1 = 0, \qquad 2y = 0 \quad \Longrightarrow \quad \left(-\frac{1}{2}, 0\right),$$

where

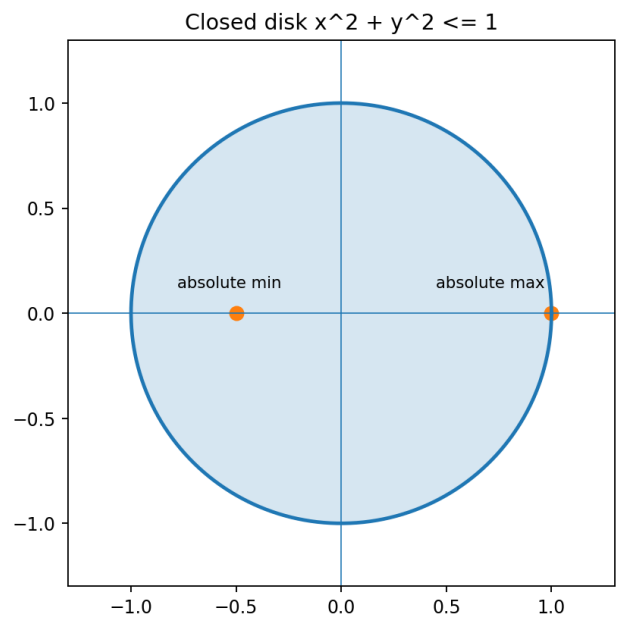
$$f = -\frac{1}{4}.$$

On the boundary, let $x = \cos\theta$, $y = \sin\theta$. Then

$$f = 1 + \cos\theta,$$

whose range is $[0, 2]$. Therefore

$$\boxed{f_{\min} = -\frac{1}{4} \text{ at } (-1/2, 0)}, \qquad \boxed{f_{\max} = 2 \text{ at } (1, 0)}.$$



22. Engineering optimisation

Suppose

$$R(x, y) = 12x + 10y - x^2 - y^2 - xy.$$

Find and classify its stationary point.

Solution

$$R_x = 12 - 2x - y, \qquad R_y = 10 - x - 2y.$$

Solving gives

$$\boxed{x = \frac{14}{3}, \qquad y = \frac{8}{3}}.$$

Now

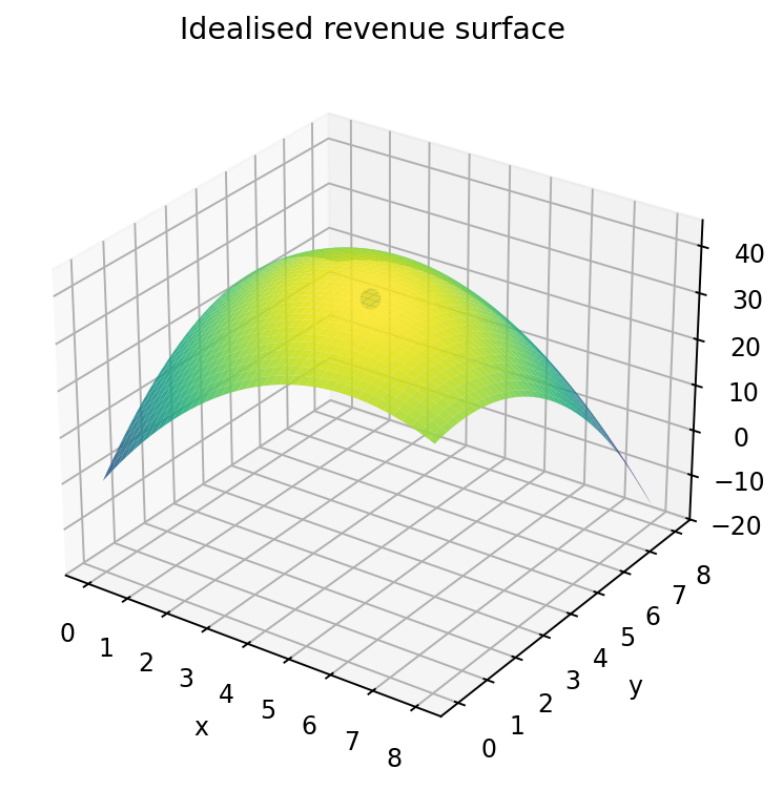
$$R_{xx} = -2, \qquad R_{yy} = -2, \qquad R_{xy} = -1,$$

so

$$D = 4 - 1 = 3 > 0, \qquad R_{xx} < 0.$$

Hence

$$\boxed{\left(\frac{14}{3}, \frac{8}{3}\right) \text{ is a local maximum}}.$$



23. Trigonometric arc length

Find the length of

$$y = \ln(\sec x), \qquad 0 \le x \le \frac{\pi}{4}.$$

Solution

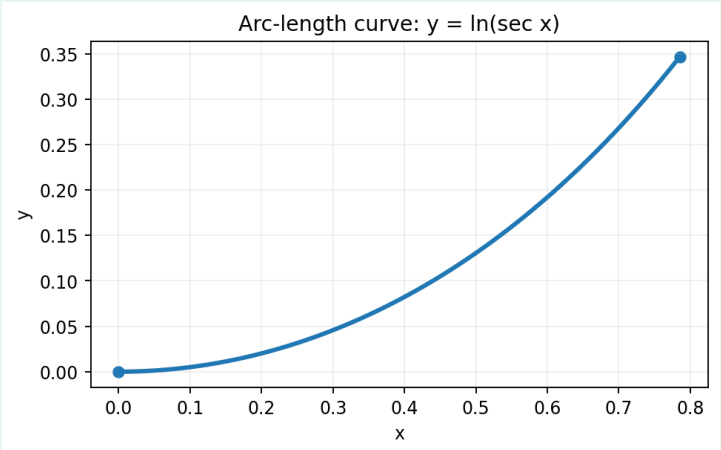
$$y' = \tan x,$$

so

$$\sqrt{1 + (y')^2} = \sqrt{1 + \tan^2 x} = \sec x.$$

Hence

$$L = \int_0^{\pi/4} \sec x \, dx = [\ln(\sec x + \tan x)]_0^{\pi/4} = \boxed{\ln(\sqrt{2} + 1)}.$$



24. Arc length using the dy form

Find the length of

$$x = \frac{y^3}{3} + \frac{1}{4y}, \qquad 1 \le y \le 2.$$

Solution

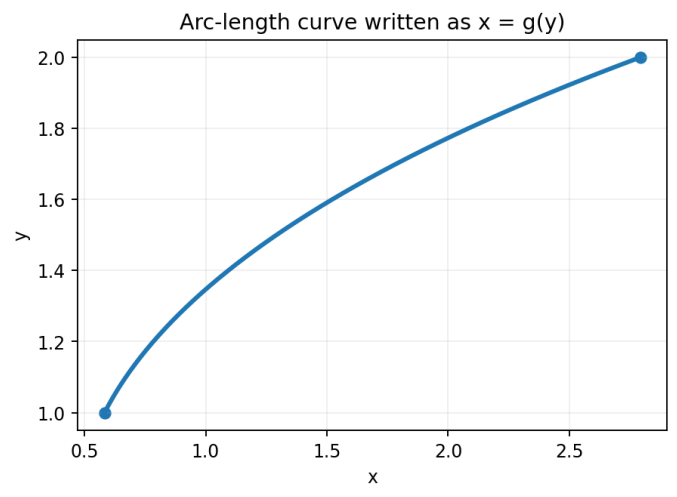
$$\frac{dx}{dy} = y^2 - \frac{1}{4y^2}.$$

Then

$$1 + \left(\frac{dx}{dy}\right)^2 = \left(y^2 + \frac{1}{4y^2}\right)^2.$$

Therefore

$$L = \int_1^2 \left(y^2 + \frac{1}{4y^2}\right) dy = \left[\frac{y^3}{3} - \frac{1}{4y}\right]_1^2 = \boxed{\frac{59}{24}}.$$



25. A different astroid segment

For

$$x^{2/3} + y^{2/3} = 16,$$

find the first-quadrant arc length from $(27, 7\sqrt{7})$ to $(64, 0)$.

Solution

Implicit differentiation gives

$$y' = -\frac{y^{1/3}}{x^{1/3}}.$$

Thus

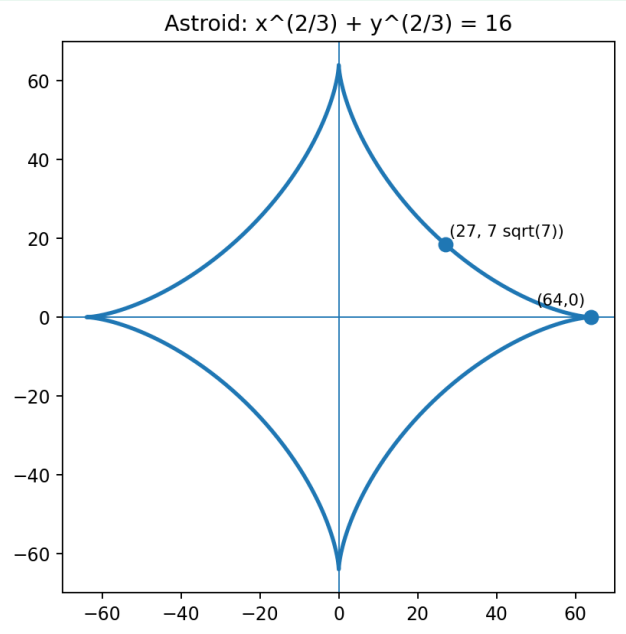
$$\sqrt{1 + (y')^2} = 4x^{-1/3}.$$

Hence

$$L = \int_{27}^{64} 4x^{-1/3}\, dx = \left[6x^{2/3}\right]_{27}^{64} = 6(16 - 9).$$

Therefore

$$\boxed{L = 42}.$$


Astroid: x^(2/3) + y^(2/3) = 16
(27, 7 sqrt(7))
(64,0)
60
40
20
0
−20
−40
−60
−60
−40
−20
0
20
40
60